

TABLE I
Clamp Meters from Different Manufacturers

Specifications		Fluke Fluke 362	Agilent U1211A	HIOKI 3288-20	Metrix+ Mod: 4250T	IDEAL 61-775
AC current	Range	200A	40A/400A/400-700A/ 700-1000A	100/1000A	1000A	0.0-999.9A
	Resolution	0.1A	0.01A/0.1 A/1A	0.1A (100A range)	1A	
	Accuracy	2% $\pm$ 5 (45-65Hz) 2.5% $\pm$ 5 (65-400Hz)	1.0 % $\pm$ 5 (400-1000A)	1.5% $\pm$ 5 digits	2% $\pm$ 5 digits	2%
DC current	Range	200A	40A/400A/1000A	100/1000A	1000A	0.0-999.9A
	Resolution	0.1A	0.01A/0.1A/1A	0.1A (100A range)	1A	
	Accuracy	2% $\pm$ 5 digits		1.5% $\pm$ 5 digits	2% $\pm$ 5 digits	2%
AC voltage	Range	600V	400V/1000V	4.2/42/420/600V	400mV-700V	0.0-750.0V
	Resolution	0.1V	0.1V	0.001V (4.2V range)	1V	
	Accuracy	1.5% $\pm$ 5 (45-400Hz)	1 % $\pm$ 5 digits	2.3% $\pm$ 8 digits	1.5% $\pm$ 5 digits	1%
DC voltage	Range	600V	400V	420mV/4.2/42/420/600V	400mV-1000V	0.0-999.9V
	Resolution	0.1V	0.1 V	0.1mV (420mV range)	1V	
	Accuracy	1% $\pm$ 5 digits	0.5% $\pm$ 5 digits	1.3% $\pm$ 4 digits	1% $\pm$ 3 digits	1%
Resistance	Range	300/3000 Ω	400 Ω / 4k Ω	420-41.99M Ω	400 Ω -40M Ω	0.0-999.9 Ohm
	Resolution	0.1/1 Ω	0.1 Ω /0.001k Ω	0.1 Ω (420 Ω range)	10k Ω	
	Accuracy	1% $\pm$ 5 digits	0.5% $\pm$ 3 digits	2% $\pm$ 4 digits	1.5% $\pm$ 5 digits	2%
Continuity		$\leq$ 70 Ω	Yes	Yes	Yes	Yes
Dimensions		205 $\times$ 60 $\times$ 22mm	106 $\times$ 273 $\times$ 43mm	57 $\times$ 180 $\times$ 16mm	249 $\times$ 89 $\times$ 38mm	270 $\times$ 103 $\times$ 48.5mm
Clamp opening		18mm	52mm	35mm diameter	52mm	51mm
Weight (gm)		196	625	150	360	500 (including battery)
Safety rating		CAT III 600V	CAT III 1000V/CAT IV 600V	CAT III 600V (current) CAT III 300V (voltage)	IEC 61010-1	CAT IV, CAT III

Insulation resistance tester

Eighty per cent of electrical maintenance and testing involves evaluating insulation integrity. Harsh installation environments, especially those with extreme temperatures and/or chemical contamination, cause deterioration to the insulations and, as a result, personnel safety and power reliability can suffer. So it is important to identify this deterioration as quickly as possible so you can take the necessary corrective



Fig. 2: Insulation resistance meter

measures. Fig. 2 shows an insulation resistance meter.

To test the insulation, we apply a highly regulated, stabilised DC voltage across a dielectric and measure the amount of current flowing through that dielectric, and then calculate the insulation resistance using Ohm's law. The current here is the leakage current and the resistance is in mega ohms. This value of resistance is used to evaluate the integrity of the conductor.

Generally, positive and negative leads are put across an insulation barrier. A third lead, which connects to a guard terminal, may or may not be available with every tester. After the connections are made, test voltage is applied for a minute, which is a standard industry parameter that allows you to make relatively accurate comparisons of readings from past tests done by other technicians. During this

interval, the resistance reading should drop or remain relatively steady.

Larger insulation systems will show a steady decrease, whereas smaller systems will remain steady because the capacitive and absorption currents drop to zero faster than on larger systems. After one minute, you should read and record the resistance value. Table II shows insulation resistance testers from different manufacturers.

Earth ground tester

A proper grounding system is of huge importance in the industrial and home environment. Proper grounding of metallic enclosures and support structures that are part of the electrical system helps avoid fatal shocks due to any kind of electrical system failure. The grounding system also provides protection against static electricity from tribocharging. Another important